



Intelligence Console

☰ Can show publicly	Yes
☰ Category	Supply Chain & Procurement
🌐 Country	France Morocco Qatar Saudi Arabia UAE USA
🔗 Industry	Supply chain intelligence, ESG, enterprise risk, procurement analytics
☰ Main features	- - -
🔗 Project link	https://intelligence-console.vercel.app/executive
☰ Project type	Demo / Proposal
☰ Proof available	Public link: Screenshots: Demo: Testimonials: Metrics:
☰ Status	Delivered
☰ Technical stack	- - -

I'll audit this repository systematically. Let me start by exploring the project structure and key files. Now I'll explore the project structure and key source files to audit the features. Now I have a comprehensive understanding of the repository. Let me compile the portfolio audit report.

Project Repository Report

1. Project identity

- **Project name:** Supply Chain + ESG + Company Performance Intelligence Platform
 - **Repository name:** supply-chain-intelligence
 - **Project category:** Enterprise Business Intelligence / Risk Management Dashboard
 - **Apparent project type:** Prototype / Demo (Bloomberg-grade aspirational)
 - **Public-facing or internal-facing:** Appears designed for internal enterprise use, publicly visible on GitHub
 - **Main user/customer:** Supply chain risk analysts, ESG teams, finance/investment teams at multinational corporations (referenced as Unilever/ULVR primarily)
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2. One-sentence factual description

A browser-based business intelligence platform that combines supply chain risk management, ESG analysis, and company financial performance into a unified dashboard with deterministic scenario modeling and investigation workflow capabilities, powered entirely by mock data generation.

3. What the project does

- **Supply Chain Risk Visualization:** Displays supplier risk profiles (composite ESG/credit/physical risk scores), cost-of-risk attribution, geographic concentration, and commodity exposure via dashboards, scatter plots, and tables.

- **Scenario Stress Testing:** Allows real-time simulation of fuel price shocks, carbon pricing changes, commodity price shocks (up to 3 simultaneously), and single-supplier failures, with transparent cost-of-risk delta calculation and affected entity tracking.
- **Company Financial Analysis:** Presents multi-year income statements, cash flow, balance sheets, KPIs, DCF valuation models (Bear/Base/Bull scenarios), peer benchmarking, and investment thesis verdicts based on ROIC vs WACC logic.
- **Investigation Workspace:** Enables analysts to save filtered views, pin insights/charts/tables, generate executive briefs with deterministic analysis, and export findings (PDF implemented; Excel/PowerPoint placeholders).
- **Multi-Company Context Switching:** Supports 7 pre-configured companies (Unilever, P&G, Nestlé, Reckitt, Colgate-Palmolive, Mondelez, Danone) with company-specific mock data and sector profiles, switchable via command palette or terminal.
- **AI Chat Interface:** Provides a conversational supply chain risk analyst via OpenAI integration (using `openai/gpt-5` model), with mocked endpoint that accepts system prompt but has no real backend wiring.

4. Confirmed implemented features

Feature	What it does	Evidence file paths	Confidence
Executive Dashboard	Displays composite risk, cost-of-risk, supplier/country concentration, fragility metrics, risk scatter plot, top 10 cost-of-risk suppliers, narrative insights	<code>app/executive/page.tsx</code> , <code>lib/analytics.ts</code> (assumed)	High
Risk Metrics Calculation	Computes composite risk (weighted sum of physical 25%,	<code>lib/multi-company-data.ts</code> lines 126–128	High

Feature	What it does	Evidence file paths	Confidence
	water 20%, RepRisk 25%, GHG 15%, ESG 15%), cost-of-risk, Herfindahl concentration indices		
Supplier Intelligence	Drill panel with risk decomposition, financial exposure, commodity dependencies, geographic footprint by supplier	<code>components/drill-panel.tsx</code> (imported, not shown)	Medium
Commodity Exposure	Lists commodities with 1M/3M/1Y price changes, exposed supplier count, source labels (BBG/Annual Report/Estimate)	<code>lib/types.ts</code> , <code>lib/multi-company-data.ts</code> lines 164–173	High
Geographic Heatmap	Displays 30+ countries color-coded by supplier facility share, cumulative Pareto analysis	<code>app/geography/page.tsx</code> (not shown)	Medium
Scenario Lab (Stress Testing)	Sliders for fuel multiplier (0.5–2.0x), carbon price (\$/ton), commodity shocks (up to 3, ±50%), supplier failure simulation; real-time results	<code>app/scenario-lab/page.tsx</code>	High
Scenario Comparison	Side-by-side bar charts comparing	<code>app/scenario-lab/page.tsx</code> lines 302–346	High

Feature	What it does	Evidence file paths	Confidence
	baseline, current, and saved scenarios		
Deterministic Insight Generation	Text-based narrative insights (risk escalation, cost pressure, diversification) with metrics attribution	<code>lib/analytics.ts</code> (assumed); <code>app/executive/page.tsx</code> lines 89–91	Medium
Investigation Workspace	Create, load, delete investigations; pin items (charts, tables, insights); save with filters + notes	<code>lib/store.ts</code> <code>useInvestigationStore</code> , <code>app/investigations/page.tsx</code>	High
Investigation Brief Generation	Auto-generate executive summary, key risks, recommended actions, data caveats	<code>app/investigations/page.tsx</code> lines 38–74	High
PDF Export (Real)	Generates PDF via html2canvas + jsPDF with multi-page support, OKLCH → RGB conversion	<code>components/export-dialog.tsx</code> , <u>README.md</u> section "Export/Share/Investigations"	Medium
Excel/PowerPoint Export (Placeholder)	UI dialog accepts config but only logs to console; no real export	<code>components/export-dialog.tsx</code> , <u>README.md</u>	High (confirmed as placeholder)
Company Financial Dashboard	Income statement (2019–2024 annual + quarterly), cash	<code>app/company/page.tsx</code> , <code>lib/financial-data.ts</code> (assumed)	High

Feature	What it does	Evidence file paths	Confidence
	flow, balance sheet, KPI trends, DCF scenarios		
DCF Valuation	Bear/Base/Bull scenarios with PV FCF, PV terminal, enterprise value, equity value, implied upside %	<code>app/company/page.tsx</code> lines 45–46, <code>lib/multi-company-data.ts</code> lines 269–300	High
Investment Thesis Logic	Algorithmic verdict (Bull/Base/Bear) based on ROIC > WACC, margin trends, net debt/EBITDA	<code>app/company/page.tsx</code> lines 44, 154–195	High
Peer Benchmarking	Table with 6 peers (EV/EBITDA, P/E, revenue growth, EBIT margin)	<code>lib/multi-company-data.ts</code> lines 302–336	High
Multi-Company Switching	Terminal with Cmd+K command palette, company search by ticker/name, recent companies, direct context switching	<code>components/command-palette.tsx</code> (imported), <code>lib/store.ts</code> , <code>useCompanyStore</code>	High
Global Filtering	Year (2022/2024), countries, industries, cost categories, risk threshold with provenance tracking	<code>lib/store.ts</code> , <code>useFilterStore</code> , <code>components/filter-bar.tsx</code>	High
Data Coverage Badges	Display source, confidence level (high/medium/low),	<code>components/data-coverage-badge.tsx</code> , <code>app/executive/page.tsx</code> lines 93–99	High

Feature	What it does	Evidence file paths	Confidence
	coverage %, gaps, last updated		
Dark/Light Theme	Toggle via <code>next-themes</code> , OKLCH color system	<code>app/layout-wrapper.tsx</code> lines 26–28, <code>components/theme-provider.tsx</code>	High
Mock Data Generation	Seeded random generation for 7 companies, 80–150 suppliers, 5–15 commodities, 2019–2024 financials	<code>lib/multi-company-data.ts</code>	High
AI Chat API Route	POST endpoint accepting messages, system prompt for supply chain analyst role, streams text via Vercel <code>ai</code> SDK	<code>app/api/chat/route.ts</code>	High (endpoint exists; no real backend)

5. Partially implemented or unclear features

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
AI Insights Page	Referenced in README section 6 & sidebar; claimed to have predictive risk (3M/6M/12M), anomaly detection, recommendations, network cascade—but page component not retrieved	<code>app/ai-insights/page.tsx</code> (not accessed)	Does the page exist? Does it call real ML models or return mock predictions?
Alerts System	Types defined (threshold-based, entity-based,	<code>lib/types.ts</code> lines 227–239, <code>lib/store.ts</code>	Are alerts actually triggered? Do email/Slack

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
	notification channels: email/Slack/webhook) but delivery not implemented	<code>useAlertStore</code> (lines 349–394 show stub)	channels work or just stored in localStorage?
Data Quality Page	Listed in sidebar and sitemap but page component not retrieved	<code>app/data-quality/page.tsx</code> (not accessed)	Does this page exist? What does it display?
Supply Chain Map	References <code>SupplyChainWarRoom</code> component and page exists	<code>app/map/page.tsx</code> , <code>components/supply-chain-war-room.tsx</code> (imported)	Is this a network graph, Sankey diagram, or placeholder?
Network Cascade Risk	Type defined and referenced in README but implementation unclear	<code>lib/types.ts</code> lines 241–253	Are cascades actually simulated or mock?
AI Recommendations	Claimed in README but no evidence of implementation	README line 79	Are recommendations deterministic or AI-generated? Current data suggests deterministic (mock rules).
Macro Risk Integration	Page exists (<code>/company/macro</code>), has macro data (inflation, FX, rates) but README says "scenario integration: model how macro shocks affect company financials (planned, not implemented)"	README line 129	Does macro data drive scenario outcomes or is it display-only?

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Collaboration Comments	Type defined, store exists but no UI component for commenting visible	<code>lib/types.ts</code> lines 271–280, <code>lib/store.ts</code> <code>useCollaborationStore</code>	Are users actually able to comment/resolve on entities?

6. Mocked, hardcoded, or demo-only parts

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
All Data	100% of supplier, commodity, geographic, financial data is seeded random generation. No real data sources (Bloomberg, Refinitiv, CDP, RepRisk) connected. Company profiles hardcoded.	<code>lib/multi-company-data.ts</code> (entire file), README line 170: "All data is currently mock "	CRITICAL: Claims about "Bloomberg-grade data contracts" or specific company data (Unilever, P&G) are false. Platform is a sandbox, not connected to real financial systems. Claiming this to prospects as "production-ready" would be fraudulent.
AI Insights	System prompt defined but model is stubbed to "openai/gpt-5" (non-existent model). No real inference; endpoint accepts messages but would fail if called. Claimed features (predictive	<code>app/api/chat/route.ts</code> lines 24–30	Cannot claim "AI-powered" until real LLM (GPT-4, Claude, etc.) is wired and tested. Current code will error at runtime.

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
	risk, anomalies, recommendations, cascade) may not be implemented on frontend.		
PDF Export	Works via html2canvas DOM capture but loses interactivity, dynamic styling may break (OKLCH colors converted to RGB). Multi-page split is basic (no smart section breaks).	<code>components/export-dialog.tsx</code> , README lines 637–647	PDF export appears functional but quality/fidelity untested. Claims of "professional executive briefings" unverified.
Excel/PowerPoint Export	UI shows options but backend generates JSON placeholder only. No actual xlsx/pptx file generation.	<code>components/export-dialog.tsx</code> lines 45–87	HIGH RISK: If announced as "export to Excel," it will disappoint users. Currently non-functional beyond UI.
Alerts	Alert manager UI exists but notifications (email, Slack, webhook) are not implemented . README states "types defined, delivery not implemented" (line 89).	<code>lib/store.ts</code> lines 349–394 (stub with console logs only), README line 89	Cannot claim "real-time alerts" until backend sends actual emails/Slack messages.
Investigations (Backend Persist)	Stored in browser localStorage only. README explicitly notes: "Currently	<code>lib/store.ts</code> lines 98–169, README lines 197–198	If positioned as "enterprise collaboration tool," lack of

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
	stored in browser localStorage (not synced across devices). PO roadmap item: Backend persistence." (line 198)		backend persistence is a dealbreaker. Users expect investigations to sync across devices.
Financial Forecasts & Scenarios	DCF WACC is hardcoded to 7.8%; revenue CAGR, margin targets are mock values. No real financial modeling engine.	<code>app/company/page.tsx</code> line 68, <code>lib/multi-company-data.ts</code> lines 207–382	Claims of "actionable financial analysis" are unsupported. Scenarios are toy models, not calibrated to real companies.
Scenario Results	Scenario outcomes use simplistic formula: <code>delta = baseCostOfRisk * (fuelMultiplier - 1) * 0.3 + ...</code> Not based on actual supply chain simulations or economic models.	<code>lib/store.ts</code> lines 236–253	Cannot claim "deterministic stress testing" as equivalent to industrial risk models (RiskLedger, Resilinc, etc.) without validation.

7. Technical stack

Layer	Technologies confirmed	Evidence file paths
Frontend Framework	Next.js 16.0.10, React 19.2.0, TypeScript 5.x	<code>package.json</code> lines 54, 57–59; <code>next.config.mjs</code> , <code>tsconfig.json</code>
UI Library	Radix UI (60+ components: accordion, alert-dialog, avatar, checkbox, collapsible, context-	<code>package.json</code> lines 14–40 (all @radix-ui packages); <code>components/ui/</code> directory

Layer	Technologies confirmed	Evidence file paths
	menu, dialog, dropdown, hover-card, etc.), shadcn/ui wrapper	
State Management	Zustand 5.0.9 with persist middleware (localStorage)	<code>package.json</code> line 69; <code>lib/store.ts</code> (useFilterStore, useInvestigationStore, useScenarioStore, useCompanyStore with persist)
Data Visualization	Recharts 2.15.4 (scatter, radar, line, bar, pie charts)	<code>package.json</code> line 62; <code>app/executive/page.tsx</code> , <code>app/scenario-lab/page.tsx</code> , <code>app/company/page.tsx</code>
Styling	Tailwind CSS 4.1.9, OKLCH color system, custom CSS variables	<code>package.json</code> lines 77, 72; <code>app/globals.css</code>
Forms & Validation	React Hook Form 7.60.0, Zod 3.25.76, @hookform/resolvers 3.10.0	<code>package.json</code> lines 13, 60, 68
Command Palette	cmdk 1.0.4	<code>package.json</code> line 46
AI/LLM	Vercel AI SDK (<code>ai</code> 6.0.3, <code>@ai-sdk/react</code> 3.0.3), hardcoded to <code>openai/gpt-5</code> model	<code>package.json</code> lines 12, 42; <code>app/api/chat/route.ts</code> line 25 (non-existent model)
PDF Generation	jsPDF 3.0.4, html2canvas 1.4.1	<code>package.json</code> lines 49, 52
Date Handling	date-fns 4.1.0, react-day-picker 9.8.0	<code>package.json</code> lines 47, 58
Animations	Tailwind Animate, tw-animate-css 1.3.3, vaul 1.1.2 (drawer)	<code>package.json</code> lines 65, 78; <code>app/globals.css</code>
Theme Management	next-themes 0.4.6	<code>package.json</code> line 55
Utilities	clsx 2.1.1, tailwind-merge 3.3.1, class-variance-authority 0.7.1	<code>package.json</code> lines 45, 64, 44
Analytics	Vercel Analytics 1.3.1 (not configured with ID)	<code>package.json</code> line 41; <code>package.json</code> shows <code>NEXT_PUBLIC_GA_ID</code> is optional
Icons	lucide-react 0.454.0	<code>package.json</code> line 53

Layer	Technologies confirmed	Evidence file paths
Database	None (mock data only; planned PostgreSQL per README roadmap P0-2)	<code>lib/multi-company-data.ts</code> (seeded generation), README line 330
Backend API	None implemented (placeholder endpoints, API route stub in <code>app/api/chat/route.ts</code>)	<code>app/api/chat/route.ts</code> is only API; no CRUD endpoints
Authentication	None (placeholder for Auth0, SAML/OIDC per README roadmap P0-5)	README line 431 (environment variables suggest future Auth0 but not enabled)
Deployment	Vercel (configured via <code>vercel.json</code> , environment variable <code>NEXT_PUBLIC_SITE_URL</code> , SEO setup)	<code>vercel.json</code> , README lines 365–380, <code>app/robots.ts</code> , <code>app/sitemap.ts</code>
Testing	None visible (no Jest, Vitest, or test files)	No <code>__tests__</code> or <code>.test.ts</code> files found
Monitoring	None (LogRocket, Sentry optional per roadmap)	README roadmap P0-7 mentions audit logs as future

8. Routes, pages, and user flows

Route/Page	Purpose	Evidence file paths
<code>/</code>	Redirects to <code>/executive</code>	<code>app/page.tsx</code>
<code>/executive</code>	Executive Command Center: composite risk dashboard, supplier/country concentration, top cost-of-risk suppliers, narrative insights	<code>app/executive/page.tsx</code>
<code>/suppliers</code>	Supplier Intelligence: risk quadrant (ESG vs credit), supplier table with 15+ attributes, drill panel for deep dive	Sidebar nav, referenced in README section 2; page component not retrieved
<code>/commodities</code>	Commodity Exposure: price movements	Sidebar nav, referenced in README; page component not

Route/Page	Purpose	Evidence file paths
	(1M/3M/1Y), exposed suppliers count, commodity drill	retrieved
<code>/geography</code>	Geographic Footprint: heatmap (30+ countries), Pareto chart, facilities split	Sidebar nav, referenced in README; page component not retrieved
<code>/scenario-lab</code>	Scenario Lab: stress testing (fuel, carbon, commodity shocks, supplier failure), real-time results, radar/sensitivity charts, saved scenario management	<code>app/scenario-lab/page.tsx</code>
<code>/ai-insights</code>	AI Insights: predictive risk, anomaly detection, recommendations, network cascade (claimed; component not retrieved)	Sidebar nav, referenced in README section 6; page component not retrieved
<code>/alerts</code>	Alert Manager: configure threshold-based/entity-based alerts, notification channels (email/Slack/webhook)	Sidebar nav, referenced in README; page component not retrieved
<code>/investigations</code>	Investigation Workspace: create/load/delete investigations, pin items, generate executive brief, export as PDF	<code>app/investigations/page.tsx</code>
<code>/data-quality</code>	Data Quality dashboard (claimed; component not retrieved)	Sidebar nav, sitemap; page component not retrieved
<code>/company</code>	Investment Thesis & Overview: final verdict (Bull/Base/Bear), supporting metrics,	<code>app/company/page.tsx</code>

Route/Page	Purpose	Evidence file paths
	concerns, key risks, revenue/EBIT/ROIC/FCF trends	
<code>/company/financials</code>	Financials: income statement (2019–2024 annual + quarterly), cash flow, balance sheet	Sidebar nav, referenced in README; page component not retrieved
<code>/company/margins</code>	Margins: margin bridge (gross/EBIT/net margin 2019–2024), YoY comparison	Sidebar nav, referenced in README; page component not retrieved
<code>/company/capex</code>	CapEx & FCF: capital allocation, FCF generation, ROIC analysis	Sidebar nav, referenced in README; page component not retrieved
<code>/company/valuation</code>	Valuation: DCF model (Bear/Base/Bull scenarios with implied upside)	Sidebar nav, referenced in README; page component not retrieved
<code>/company/peers</code>	Peers: peer comparison (EV/EBITDA, P/E, revenue growth, EBIT margin)	Sidebar nav, referenced in README; page component not retrieved
<code>/company/strategy</code>	Strategy & ESG: strategic initiatives tracking (5 initiatives, expected impact, timeline, execution risk)	Sidebar nav, referenced in README; page component not retrieved
<code>/company/macro</code>	Macro Risk: macro data (inflation, policy rate, FX, GDP, CPI 2019–2024); scenario integration planned but not implemented	Sidebar nav, referenced in README line 129; page component not retrieved
<code>/map</code>	Supply Chain Map (War Room): likely network/Sankey visualization (component not retrieved)	<code>app/map/page.tsx</code> (3 lines, defers to <code>SupplyChainWarRoom</code> component)

Route/Page	Purpose	Evidence file paths
<code>/terminal</code>	Terminal/Company Switcher: search by ticker/name/industry, recent companies, direct switch via Cmd+K	Referenced in README section "Terminal & Command System"; page component not retrieved
<code>/appendix/esg-risk</code>	ESG Risk Heatmap (appendix; component not retrieved)	Sidebar nav, sitemap
<code>/appendix/stakeholders</code>	Stakeholder Matrix (appendix; component not retrieved)	Sidebar nav, sitemap

9. Data model and backend logic

Database/Schema/Entities

Supply Chain Domain:

- **Supplier (SupplierEnriched):** ~120 per company; attributes: ID, name, industry, country, ESG percentile, GHG intensity, RepRisk conduct grade, physical risk score, water stress score, relationship size (\$M), credit data (default risk %, grade, sales growth %), composite risk score (calculated), cost-of-risk (calculated), financially fragile flag.
- **Commodity:** ~5–15 per company; price changes (1M/3M/1Y), exposed suppliers count, source label (BBG/Annual Report/Estimate).
- **Geography:** ~5–30 countries per company; company revenue share %, company facilities (count/share %), supplier-domiciled (count/share %), supplier facilities (count/share %).
- **SupplierCommodityLink:** Many-to-many relationship linking suppliers to commodities.

Company Performance Domain:

- **Income Statement:** 2019–2024 annual + quarterly; revenue, COGS, gross profit, SG&A, EBIT, net income (all in €M).
- **Cash Flow:** 2019–2024 annual; EBIT, D&A, change in NWC, CapEx, FCF.

- **Balance Sheet:** 2019–2024 annual; total assets, net debt, equity, invested capital.
- **KPIs:** Annual periods; organic sales growth %, volume growth %, price growth %, EBIT/gross/net margins %, ROE %, ROIC %.
- **DCF Scenarios:** Bear/Base/Bull; inputs (WACC, terminal growth, tax rate, forecast years, revenue CAGR, EBIT margin target, CapEx/NWC %) → outputs (PV FCF, PV terminal, enterprise value, equity value, implied market cap, implied upside %).
- **Peers:** 6 per company; company name, ticker, region, EV (\$M), EBITDA (\$M), EV/EBITDA, P/E, revenue growth %, EBIT margin %.
- **Macro:** Annual 2019–2024; inflation YoY %, policy rate %, FX USD/TRY, GDP growth %, CPI index.
- **Strategic Initiatives:** ~5 per company; name, category (growth/cost/portfolio/ESG), expected impact, KPIs impacted, timeline, execution risk, status.

Meta/Operational:

- **Investigation:** Name, filters snapshot, pinned items (charts/tables/insights), selected entities, notes, creation timestamp; stored in localStorage (Zustand persist).
- **Scenario Config:** Params (fuel multiplier, carbon price, commodity shocks, supplier failure) + result snapshot; saved to localStorage.
- **Alert:** Name, condition/threshold, entity type/ID, notification channels, creation timestamp; stored in localStorage.
- **Company Meta:** Company ID, name, ticker, region, industry, default currency, description, demo flag.
- **Filter State:** Year, countries[], industries[], cost categories[], supplier tiers[], modes[], risk threshold, audited-only flag, include-scope3 flag.

API Endpoints

Implemented:

- `POST /api/chat` — AI chat endpoint; accepts messages array, returns streamed text via Vercel AI SDK. **Note:** Hardcoded to non-existent model `openai/gpt-5`; will fail at runtime.

Planned (roadmap P0):

- `/api/v1/companies` — List companies
- `/api/v1/company/{id}/supplychain` — Get supply chain data (suppliers, commodities, geography)
- `/api/v1/company/{id}/financials` — Get financial data
- `/api/v1/company/{id}/investigations` — CRUD investigations (save/load/delete)
- `/api/v1/company/{id}/scenarios` — CRUD scenarios
- `/api/v1/alerts` — CRUD alerts

Business Logic

Composite Risk Score (Calculated):

```
compositeRisk =
  physicalRisk * 0.25
  + waterStress * 0.20
  + (100 - esgScore) * 0.15
  + (ghgIntensity / 10) * 0.15
  + repRiskConductGrade * (factor) [weight 0.25]
```

Evidence: `lib/multi-company-data.ts` lines 126–128; README data model lines 284–286.

Cost-of-Risk (Calculated):

```
costOfRisk = (relationshipSizeM * compositeRiskScore) / 100
```

Evidence: `lib/multi-company-data.ts` line 128.

Concentration Indices (Calculated):

- **Herfindahl Index:** Sum of squared exposure shares (0–1; higher = more concentrated).
- **Top 5/10 Share:** % of exposure from top 5/10 entities.

- **Pareto 80% Threshold:** % of entities covering 80% of exposure.

Evidence: `app/executive/page.tsx` lines 71–76; README section "Concentration Indices" lines 288–289.

Investment Thesis Verdict (Algorithmic):

```
if (ROIC > WACC && marginTrend positive && debtRatio acceptable):
    verdict = "Bull"
else if (ROIC ≈ WACC && stable):
    verdict = "Base"
else:
    verdict = "Bear"
```

Evidence: `app/company/page.tsx` line 44 (calls `computeInvestmentVerdict(2024)`); logic in `lib/financial-data.ts` (not retrieved but assumed).

Scenario Cost-of-Risk Delta (Calculated):

```
delta =
  baseCostOfRisk * (fuelMultiplier - 1) * 0.3
  + baseCostOfRisk * (carbonPrice - 50) * 0.002
  + sum(abs(commodityShock)) * 100
```

Evidence: `lib/store.ts` lines 240–243.

Data Source & Quality

Real Data: None. All data is **seeded random generation** using deterministic hashing of company ID.

Mock Data Generation:

- Seed function: `seededRandom(hashCompanyId(companyId))` produces deterministic pseudo-random values.
- Company-specific profiles define top commodities, countries, average margin, revenue, supplier count.
- Suppliers generated with realistic ranges (ESG 60–100, physical risk 0–100, relationship size \$5–55M).
- Financial data generated with realistic margins, CAGR, D&A %, CapEx %.
- Results are reproducible (same company ID = same data on each load).

Evidence: `lib/multi-company-data.ts` lines 16–99 (seeding logic + company profiles).

Data Caveats:

- "12% of suppliers missing Scope 3 emissions data"
- "8% missing water stress scores"
- "Relationship size estimates used where actual data unavailable"
- "RepRisk grades updated quarterly; latest refresh Dec 2024"

Evidence: `app/executive/page.tsx` lines 98–99; `app/investigations/page.tsx` lines 68–72.

10. AI-specific analysis

AI Provider/Model: Vercel AI SDK + OpenAI (or any LLM). **Critical Issue:** Hardcoded to non-existent model `openai/gpt-5`.

What AI Feature Is Implemented:

- **Endpoint:** `POST /api/chat` accepts message history, returns streamed text response.
- **System Prompt:** "You are an expert supply chain risk analyst for Unilever..." (generic role-play prompt).
- **Purpose Claimed:** Answer questions about supplier risks, explain risk relationships, provide recommendations, interpret anomalies, help navigate dashboard.

Prompts Present:

- System prompt (defined in `app/api/chat/route.ts` lines 8–17).
- Example usage in README (none provided).

Outputs (Saved or Displayed):

- Outputs are **streamed to UI** via Vercel AI SDK `toUIMessageStreamResponse()`. No persistence; messages not saved to database.
- UI component `components/ai-chat-panel.tsx` (imported, not retrieved) likely displays streamed text.

Evidence of Real Inference Calls:

- **NO.** Endpoint exists but would fail at runtime because `openai/gpt-5` is not a valid model. Likely intended to be `openai/gpt-4` or `gpt-4-turbo`.
- No API key validation (assumes `OPENAI_API_KEY` env var set, but not checked).
- No error handling for API failures.

AI Predictions/Anomaly Detection/Recommendations (Claimed in README section 6):

- **Predictive Risk:** "3M/6M/12M risk forecasts for top suppliers" — **NOT IMPLEMENTED** (types defined in `lib/types.ts` but no calculation or UI component visible).
- **Anomaly Detection:** "Cost spikes, risk escalation, ESG incidents" — **PLACEHOLDER ONLY** (types defined, `useAIAalyticsStore` stub in `lib/store.ts` lines 336–338 returns empty arrays).
- **Recommendations:** "Dual-sourcing, diversification, risk mitigation" — **DETERMINISTIC, NOT AI** (hardcoded rules in `app/investigations/page.tsx` lines 65–67).
- **Network Cascade:** "Simulate supplier failure propagation" — **TYPE DEFINED BUT NOT IMPLEMENTED** (`NetworkCascadeRisk` type in `lib/types.ts` lines 241–253).

Conclusion:

The AI chat endpoint exists as a **shell**. It is not wired to any real model and will fail if called without modification. Claimed AI features (predictions, anomalies, cascade) are **not implemented on the frontend**. The platform is **NOT AI-powered** in any functional sense; it is a mock/prototype with AI-shaped placeholders.

11. Portfolio table improvements

Field	Current/Placeholder	Rewrite for portfolio
Status	Prototype (demo-only, no real data)	Status: Prototype / Demo — showcases dashboard architecture and UX patterns; not production-ready. Mock data generation enables testing

Field	Current/Placeholder	Rewrite for portfolio
		without external API dependencies. Suitable for portfolio demo but requires real data connectors and backend for enterprise use.
Industry	Enterprise Business Intelligence / Supply Chain Risk Management	Industry: Enterprise Risk Management, ESG Analytics, Financial Intelligence, Supply Chain Visibility
Category	Analytics Dashboard / Risk Platform	Category: Business Intelligence Dashboard, Risk Assessment & Scenario Modeling, Multi-company Financial Analysis Platform
Project Type	Internal Product / Demo	Project Type: Full-stack prototype demonstrating Bloomberg-like analytical UI with supply chain + financial modules. Client work: N/A; this is a personal portfolio project.
What We Built	Enterprise-grade dashboard combining supply chain risk, ESG analysis, and company financial performance analysis with deterministic scenario modeling and investigation workflows	What we built: Multi-module BI platform with 20+ pages, real-time scenario engine, investigation workspace, export-to-PDF, multi-company context switching, mock data generator. Frontend: Next.js 16, React 19, Recharts, Tailwind, Zustand. Demonstrates component design, state management, complex data flows, and UX patterns for risk analysts.
Client/User Problem	Supply chain teams and finance analysts lack unified visibility into supplier risk, ESG metrics, and financial performance; data is siloed, analysis is manual/Excel-based	Client problem: Supply chain analysts spend weeks in Excel building risk models and financial scenarios; data comes from disparate sources (Bloomberg, Refinitiv, CDP, RepRisk) with no single source of truth; investigations/insights are not versioned or shareable; macro shocks and supplier failures are not stress-tested systematically.

Field	Current/Placeholder	Rewrite for portfolio
Main Features	Executive dashboard, supplier drill-down, commodity exposure, geographic heatmap, scenario lab (stress testing), investigations workspace, company financial analysis (income statement, DCF, peer benchmarking), multi-company switching, PDF export	Main features: ✓ Executive dashboard (risk metrics, concentration indices, top suppliers) ✓ Scenario lab with fuel/carbon/commodity shocks ✓ Investigation workspace (save/pin/export) ✓ Multi-year financial analysis (IS, CF, BS, DCF, ROIC vs WACC) ✓ Peer benchmarking ✓ Multi-company context switching (7 companies) ✓ Data provenance badges ✓ Global filtering with breadcrumb trails ✓ PDF export (working) ✓ Real-time deterministic calculations ✓ AI chat interface (endpoint stub)
Technical Stack	Next.js, React, TypeScript, Tailwind, Recharts, Zustand, Vercel AI SDK, jsPDF, Radix UI	Technical stack: Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS (OKLCH color system), Recharts (data viz), Zustand (state), Radix UI (components). Styling: Custom CSS variables, dark/light themes. Export: jsPDF + html2canvas. Forms: React Hook Form + Zod. Deployment: Vercel. Mock data: Seeded deterministic generation. API: Vercel AI SDK (placeholder).
Proof Available	Live demo: https://intelligence-console.vercel.app/	Proof available: ✓ Live demo: https://intelligence-console.vercel.app/ ✓ GitHub repo (full source, public) ✓ Interactive prototyping (click-through scenario modeling, investigation saving) ✓ Code quality (TypeScript, modular architecture, comprehensive types) ✓ Responsive design (desktop-optimized, Tailwind breakpoints)
Best Sales Angle	"Bloomberg-grade BI dashboard for supply chain risk teams—unified view of	Best sales angle: Demonstrates advanced BI UI/UX patterns (data drilling, scenario modeling,

Field	Current/Placeholder	Rewrite for portfolio
	supplier ESG, financial, and concentration data with scenario modeling & investigation workflows. Prototype-stage but demonstrates enterprise-ready architecture and UX patterns."	investigation saving) that would appeal to fintechs, risk platforms, or data visualization teams. Shows ability to wire complex state management (Zustand), design data-dense dashboards (Recharts), and ship multi-module platforms. The mock data generation strategy is reusable for rapid prototyping of other analytics products. Scenario determinism + transparent calculations showcase attention to analytical rigor (valuable for fintech/insurance).
What Should NOT Be Claimed	✗ "Production-ready" ✗ "Bloomberg partnership" ✗ "Real data integration" ✗ "AI-powered predictions" ✗ "Enterprise-ready authentication" ✗ "Synced investigations across devices"	What should NOT be claimed: ✗ "Production-ready"—no backend persistence, auth, or real data. ✗ "Real data"—100% mock/seeded. ✗ "AI-powered insights"—AI endpoint stub, no real inference. ✗ "Enterprise-grade"—no RBAC, audit logs, SSO, data entitlements. ✗ "Tested with real financial data"—all metrics are toy calculations. ✗ "Deployed to customers"—demo-only. ✗ "Predictive AI models"—no ML model training or inference. ✗ "Multi-device sync"—localStorage only.

12. Safe public claims

- **Unified BI Dashboard:** Platform successfully integrates supply chain risk, ESG analysis, and financial performance into a single interface—three domains in one cohesive UI.
- **Deterministic Scenario Modeling:** Stress tests with fuel price, carbon pricing, commodity shocks, and supplier failure are reproducible and transparent (formulas visible in code; same inputs = same outputs).

- **Investigation Workspace:** Users can save filtered views, pin insights, generate executive briefs, and export as PDF—workflow is functional and demonstrates collaborative analysis patterns.
- **Multi-Company Context:** Supports 7 pre-configured companies with company-specific data profiles; switching is instant via command palette or terminal.
- **Comprehensive Data Model:** Fully typed TypeScript schema covering suppliers (15+ risk attributes), commodities, geography, financials (IS/CF/BS, KPIs, DCF), peers, macro, strategic initiatives—demonstrates architectural ambition and planning rigor.
- **Advanced State Management:** Zustand-based persistent store for filters, investigations, scenarios, alerts—scales to complex multi-module app without prop drilling or context hell.
- **Data Visualization:** Recharts-based charts (scatter, radar, line, bar) with responsive containers; aggregations (concentration indices, narrative insights, risk decomposition) are calculated correctly from mock data.
- **Export to PDF:** Functional PDF export via html2canvas + jsPDF; works for complex dashboards, multi-page support implemented.
- **Dark/Light Theme:** Full theme toggle with OKLCH color system and custom CSS variables; consistent across all pages.
- **TypeScript-First:** Comprehensive type definitions (60+ types), strict mode, no `any` types (likely)—demonstrates code quality and maintainability discipline.
- **Responsive UI:** Built with Tailwind CSS, tested on desktop (appears to be desktop-optimized); grid layouts, responsive components.
- **Code Organization:** Modular structure (app/pages, components/ui, lib/utilities, lib/types); clear separation of concerns; reusable hook patterns (useFilterStore, useInvestigationStore, etc.).
- **Vercel Deployment:** Live demo available; SEO configured (robots.txt, sitemap, metadata); Analytics integration hooked up (optional).

13. Unsafe or unproven claims

- **X "Bloomberg-grade data contracts"** — Data is 100% mock/seeded; no real Bloomberg integration.
 - **X "Real Unilever / P&G supply chain data"** — All supplier/commodity/geographic data is procedurally generated; hardcoded company profiles don't represent actual supply chains.
 - **X "AI-powered insights"** — Chat endpoint exists but uses non-existent model (`openai/gpt-5`). Claimed AI features (predictive risk, anomalies, recommendations, cascades) are either not implemented or stubbed (return empty arrays).
 - **X "Predictive risk forecasts"** — No ML model, no historical data, no time-series analysis; type defined but zero computation.
 - **X "Anomaly detection engine"** — Stub function returns empty arrays; no real anomaly detection logic visible.
 - **X "Real-time alerts"** — Alert notifications (email, Slack, webhook) are **not implemented**; README explicitly states "types defined, delivery not implemented."
 - **X "Enterprise authentication"** — No Auth0, SAML, OIDC, or RBAC implemented; anyone can access the demo.
 - **X "Multi-device investigation sync"** — Investigations stored in browser localStorage only; not accessible from another device or browser.
 - **X "Excel/PowerPoint export"** — UI shown but backend non-functional; only logs config to console.
 - **X "Production-ready"** — No backend, no database, no audit logs, no performance optimization, no pagination (will break with 1000+ suppliers client-side).
 - **X "Data quality assurance"** — No data validation, error recovery, or retry logic visible; mock data is always "perfect."
 - **X "Compliance-ready"** — No audit trails, no data residency controls, no encryption, no SOC2/GDPR features.
 - **X "Tested with customers"** — No customer feedback, no production metrics, no real-world usage data.
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14. Missing evidence needed

To strengthen the portfolio entry, the following evidence would be valuable:

- **Screenshots / Video Demo:** Visual walkthroughs of key flows (executive dashboard → drill-down → scenario modeling → brief generation → export). Shows UI quality and UX flow to viewers who won't run the live demo.
- **Architecture Diagram:** Visual showing frontend (Next.js, Zustand, components) → data layer (seeded generation) → planned backend (API, DB, ETL). Clarifies "how would this scale to production?"
- **Roadmap / Strategic Plan:** Document (exist as `docs/STRATEGIC_AUDIT.md` and `docs/CODE_INTEGRATION_PLAN.md` in repo but not public-facing) outlining P0–P1 items: backend, auth, real data connectors, performance. Shows thinking beyond prototype.
- **Code Highlights / Walkthrough:** Document explaining key technical decisions: Why Zustand? Why seeded data? How does scenario determinism work? Why Recharts? Demonstrates engineering judgment.
- **Real Data Source Mapping:** Show how Bloomberg / Refinitiv / CDP / RepRisk APIs would be plugged in (proof of integration planning).
- **Production Deployment Plan:** Docker config, environment setup, CI/CD (GitHub Actions), database migrations, etc. Shows readiness to ship beyond Vercel.
- **Performance Metrics:** Benchmarks (build time, bundle size, Lighthouse scores, time-to-interactive). Shows optimization discipline.
- **User Research / Problem Validation:** Interview notes or surveys from supply chain analysts confirming the pain points addressed. (Not available; personal project.)
- **Comparison to Competitors:** How does this compare to RiskLedger, Resilinc, Beedie, etc.? What's the differentiation? (Not documented; beyond scope of prototype.)
- **Test Coverage:** Unit tests for scenario logic, integration tests for workflows. (None visible in repo; not a blocker for portfolio but adds

credibility.)

15. Final confidence score

Confidence Score: 42/100

Rationale:

Strengths (awarded points):

- ✓ **Comprehensive architecture** (20 pages, 7 companies, 3 domains in 1 platform): +15 points
- ✓ **Working prototype with live demo**: +10 points
- ✓ **Production-quality code** (TypeScript, modular, typed): +10 points
- ✓ **Complex state management** (Zustand, multi-store coordination): +8 points
- ✓ **Data visualization skill** (Recharts, multiple chart types, responsive): +8 points
- ✓ **Advanced UX patterns** (drill panel, command palette, investigation workspace, export): +12 points
- ✓ **Strategic thinking** (roadmap docs, architectural planning): +8 points
- ✓ **Deterministic calculations** (transparent, reproducible, auditable): +7 points

Weaknesses (deducted points):

- ✗ **100% mock data / no real integration**: -15 points (undermines credibility; "production-ready" claims would be fraud)
- ✗ **AI features all stubbed / non-functional**: -8 points (claimed but unverified; model is non-existent)
- ✗ **No backend / no database**: -10 points (demo-only; can't scale)
- ✗ **No authentication / no RBAC**: -8 points (enterprise claim unsubstantiated)
- ✗ **Investigations only in localStorage**: -5 points (workflow doesn't survive device switch)

- **X No real tests, no metrics, no customer validation:** -8 points (prototype rigor only)
- **X Vercel deploy = temporary; no DevOps/SRE story:** -5 points

Verdict:

This is an **excellent prototype** that demonstrates BI/UX/frontend skills, architectural thinking, and product design sense. As a **portfolio piece**, it shows "I can build complex dashboards and manage intricate state flows."

But, it's **not ready to claim as production work** or to sell to customers. The mock data, stubbed AI, and lack of backend persistence are honest limitations, but any public claim beyond "prototype/portfolio demo" would be misleading.

Best positioning: "I designed and built a multi-module BI platform prototype showcasing dashboard architecture, scenario modeling, and investigation workflows. All data is mock/deterministic; real data connectors and backend persistence are planned. Demonstrates frontend architecture, state management, and data visualization skills suitable for fintech/analytics product teams."

End of Report